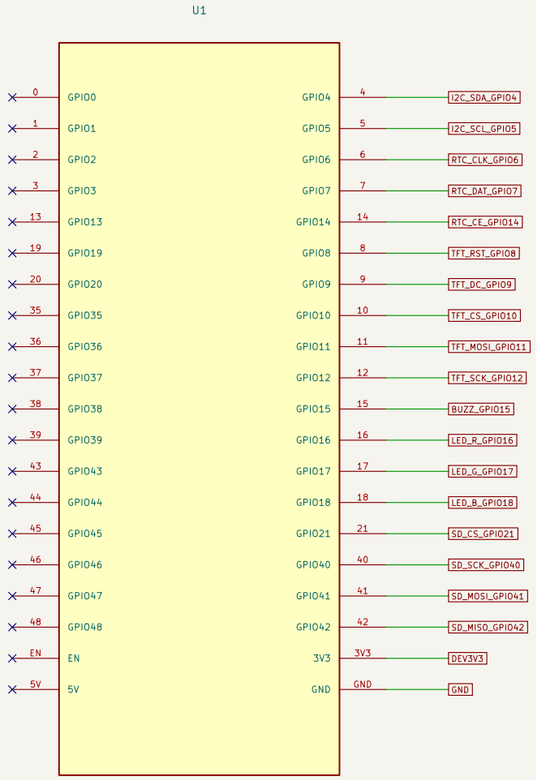
AKENA NEOGUARD | SPI DISPLAY SYSTEM INTEGRATION

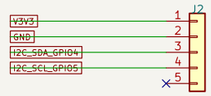
Matches current ST7789 240x280 firmware. Every used pin is wired; X marks intentionally unused external connections.

ESP32-S3 controller



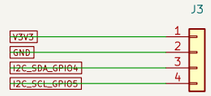
Logical GPIO identities; not header positions.
X = unused external pin (including GPIO13).
EN / USB / memory remain managed by devboard.
GPIO26–34 belong to module memory, not headers.

MAX30102 OXIMETER

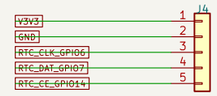


Shared I2C: SDA GPIO4, SCL GPIO5.
Sensor headers are functional interfaces.
Reorder cable wires to match purchased modules.

MLX90614 INFRARED

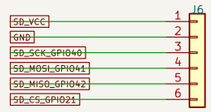


DS1302 RTC



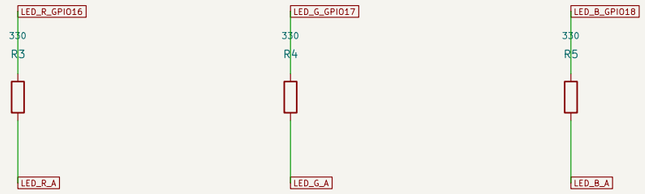
CR2032: trickle charging disabled in firmware.

SD CARD MODULE



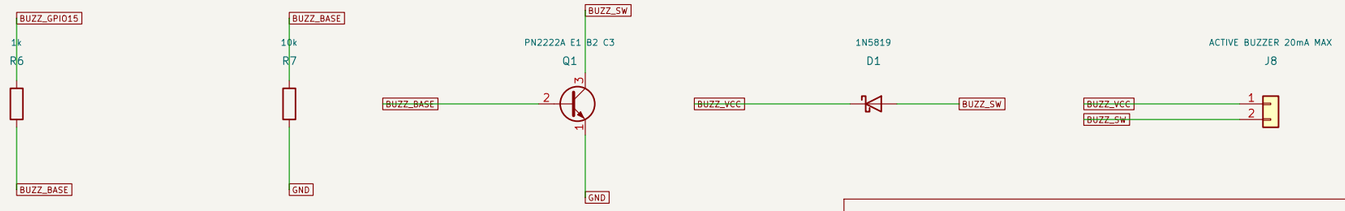
RGB COMMON CATHODE





Each LED colour has its own 330-ohm resistor.

BUZZER DRIVER | GPIO15



PN2222A E1 B2 C3; verify purchased transistor. D1 cathode -> buzzer positive. Driver sized for <=20mA.

SPI display

File: NeoGuard-display.kicad_sch

Power and controller cable

File: NeoGuard-power.kicad_sch

Display SPI GPIO8–12 -> sheet 2. GPIO13 unused. SD SPI GPIO21/40/41/42.

GPIO names are logical signals; verify actual module headers and voltages

NeoGuard research prototype

Sheet: /
File: NeoGuard.kicad_sch

Title: Akena NeoGuard – SPI System

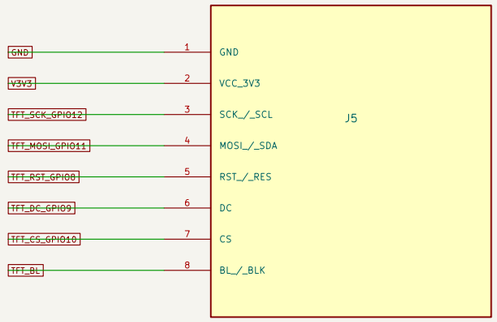
Size: A3	Date: 2026-10-29	**Rev: SPI-D**
KiCad E.D.A. 10.0.4		Id: 1/3

ST7789 240 x 280 | SPI DISPLAY CONNECTIONS

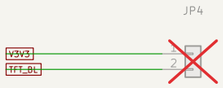
This is the tested SPI screen called "OLED" in earlier project notes. J5 is a cable header; match wires by signal name.

J5 PIN	DISPLAY SIGNAL	ESP32-S3
1	GND	Common ground
2	VCC	Selected 3.3V rail
3	SCK / SCL	GPIO12
4	MOSI / SDA	GPIO11
5	RST / RES	GPIO8
6	DC	GPIO9
7	CS	GPIO10
8	BL / BLK	JP4 backlight option

J5 / DISPLAY CABLE



BACKLIGHT OPTION



JP4 normally OPEN / DNP. Fit only for a verified 3.3V-compatible BL input with current limiting. No firmware backlight GPIO is assigned.

INTERFACE NOTES

SDA / SCL on this screen mean SPI MOSI / SCK.
Do not connect them to sensor I2C GPIO4 / GPIO5.
All signal levels are 3.3V. No LCD level shifter.
GPIO13 and J1 pin22 are intentionally unconnected.
Header order is functional: verify the actual module.

The 16x2 LCD, AHCT245 buffer, contrast trimmer and LCD backlight resistor have been removed.

GPIO names are logical signals; verify actual module headers and voltages

NeoGuard research prototype

Sheet: /SPI display/
File: NeoGuard-display.kicad_sch

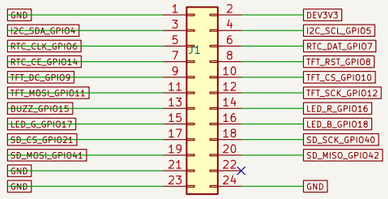
Title: Akena NeoGuard – ST7789 Display

Size: A3	Date: 2026-10-29	**Rev: SPI-D**
KiCad E.D.A. 10.0.4		Id: 2/3

POWER SELECTION | CONTROLLER CABLE

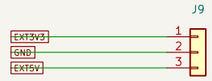
No regulator on this carrier. Connect regulated supplies only. Never short external power to a USB-powered ESP32 rail.

ESP32-S3 FUNCTIONAL CABLE



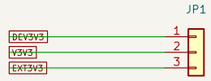
J1 is not a devboard socket.
Pin22 unused: leave unconnected.
Map cable wires by GPIO name.
USB remains on the external controller.

REGULATED INPUT



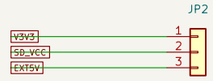
J9: 1=EXT3V3, 2=GND, 3=EXT5V
EXT5V only if SD / buzzer needs it.

3V3 SOURCE



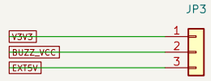
JP1: 1-2 DEV3V3; 2-3 EXT3V3
One shunt only. Do not bridge both.

SD SUPPLY



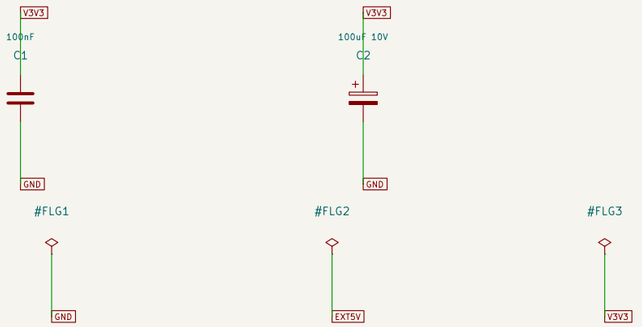
JP2: 1-2 = 3V3; 2-3 = 5V
Select to match the actual SD module.

BUZZER SUPPLY

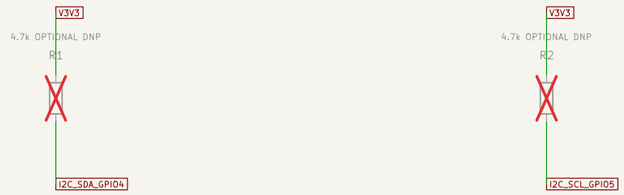


JP3: 1-2 = 3V3; 2-3 = 5V
Select to match the actual buzzer.

DECOUPLING: SENSOR 3V3 AND SD SUPPLY



OPTIONAL I2C PULL-UPS



R1/R2 DNP when breakout pull-ups already exist. SDA/SCL must be pulled up to 3.3V, never 5V.

GPIO names are logical signals; verify actual module headers and voltages

NeoGuard research prototype

Sheet: /Power and controller cable/
File: NeoGuard-power.kicad_sch

Title: Akena NeoGuard – Power and Controller Cable

Size: A3	Date: 2026-10-29	**Rev: SPI-D**
KiCad E.D.A. 10.0.4		Id: 3/3